



COMP 5009
Data Mining

Assignment Report on
**Monitoring of Vegetation Conditions on Wadjuk
Noongar Country**

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Summary

This project was to be developed with the aim of developing predictive models that would categorize the vegetation conditions in the western region of Australia Wadjuk Noongar country, in Western Australia. The data were 5,000 labeled data training set and 500 unlabeled data test set, and each row represented a land parcel or an ecological and geographic and disturbance-related features. The target classes were Healthy, Degraded, and At Risk however the classes were intentionally unequal to provide the environmental monitoring problems in real life.

It was an end-to-end machine learning pipeline where data preparation (cleaning, imputation, duplicates, scaling, encoding, and balancing classes) preceded the next step which was classification and model tuning and finally prediction of the test data.

The four classifiers, which have been tried, are k-Nearest Neighbors (k-NN), Naive Bayes (NB), Decision Tree (DT), and Random Forest (RF). The most successful models were random Forest and k-NN which were achieved after a series of attempts and trials. These models were retrained using the whole training set and then they were used to make predictions using the test set and consequential outputs were in the desired format of a CSV file.

The results showed that random forest validation accuracy was 0.906 and macro-F 1 was 0.816 and k-NN validation accuracy was 0.863 and macro-F 1 was 0.789. These results suggest that the ensemble-based and the distance-based algorithms can be implemented correctly to the data and that Naive Bayes and Decision Tree demonstrated lower results.

Methodology

Data Preparation

Data preparation is also very essential in the actual world predictive modeling since raw data is usually ill-formatted and it can negatively affect the performance severely. These were the issues that were identified and addressed in a systematic way.

i. Irrelevant attributes

The data had fields that were similar to IDs and were used as key fields. These properties do not have predictive contents and their existence can cause models to memorise and not to generalise. This is the reason why these attributes were removed in the training and test datas to ensure that only features that made sense were used to carry out the learning. This is according to the best practice of data mining in which non-predictive keys are not modeled.

ii. Missing entries

The dataset had gaps of numeric and categorical data. The values that are missing would have been disregarded that would have reduced the training data and this could have affected the model. Instead:

- Numerical items were entered in with their median, which is not susceptible to skewness and outliers.
- Mode (most frequent value) was used to fill the categorical values. This plan does not alter the information integrity, and produces no artificial synthetics, and it is strongly recommended in the process of predictive modelling (Kuhn & Johnson, 2013).

iii. Duplicate records

It was discovered that the training set had duplicate rows. Cases which are replicated would also disrupt the training process as the training would be biased towards the replicated cases as opposed to the other cases. They were sorted in a manner that a single row had one single observation.

iv. Data type corrections

Intrinsic categorical aspects were coded in numbers. In the case where they are not transformed, models might misinterpret them as ordered quantities. These are explicitly coded to categorical and one-hot to ensure that they are appropriately coded.

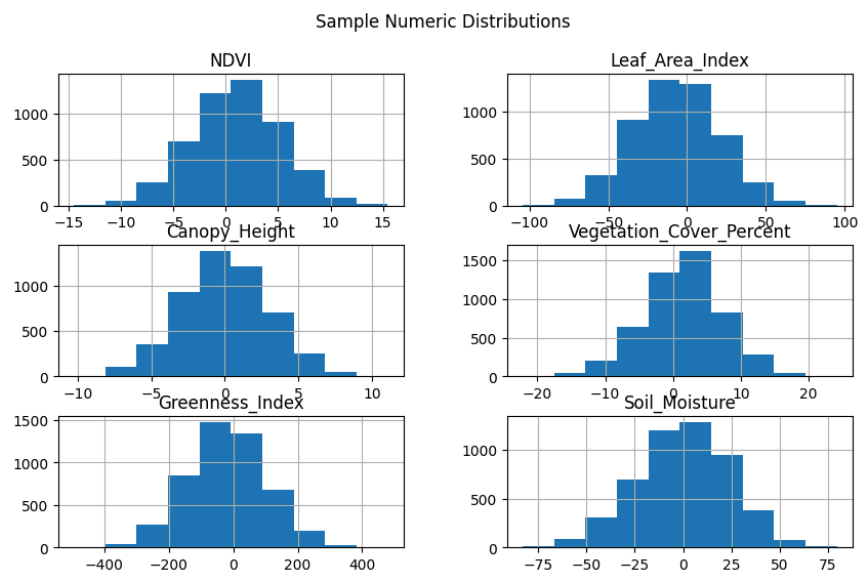


Figure 1: Sample Numeric Distributions

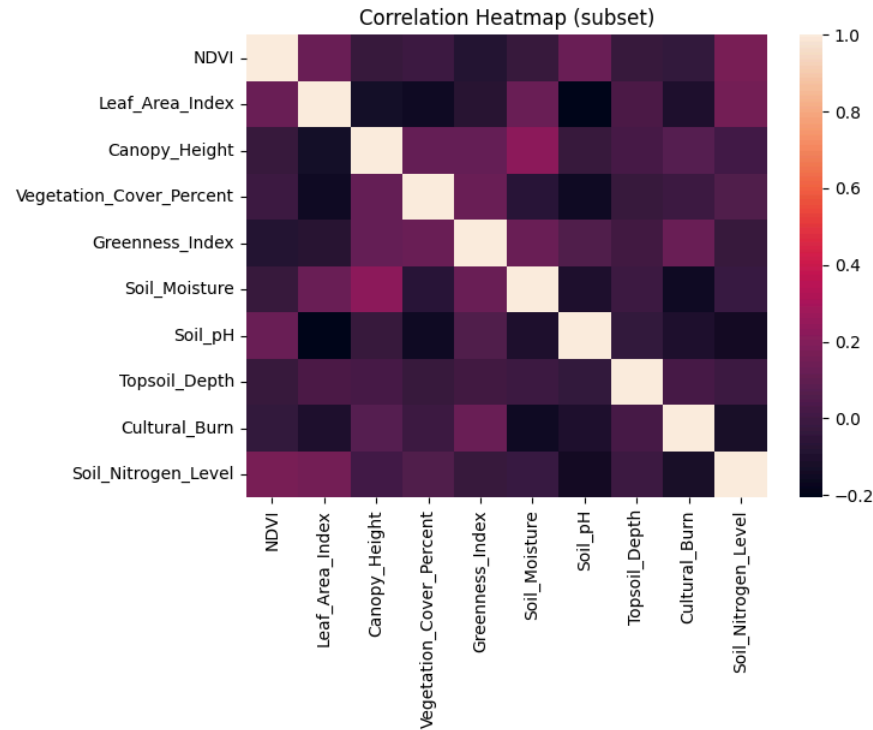


Figure 2: Correlation Heatmap

v. Data transformation (scaling and standardization)

The variables included in the data set were measured in scales that were highly different (e.g. soil chemistry vs. distance to water). The distance-based approaches such as the k-NN are highly prone to feature scaling whereby large scale variables crowd out the computation of the Euclidean distance. To solve it, StandardScaler was applied, and the features were realigned to form features that had a zero mean and unit variance. Scaling enhances comparability between heterogeneous features and also stabilizes optimization in scalability of models such as k-NN and Naive Bayes.

vi. Class imbalance

Among the most significant ones was the problem of the lack of equality in classes: the Degraded class was the most numerous, and the At Risk underrepresentation occupied the minority. Such distorted data will be affected by training so that the model results will be biased to prediction of majority classes. To mitigate this, random oversampling was done in the training split which ensured an equal distribution of classes. Even though oversampling increases susceptibility of the model to minority classes, the model can artificially increase cross-validation. This is a disadvantage that has been cited in the imbalance learning studies (Chawla et al., 2002).

vii. Outliers

There were drastic environmental gauge measurements. Instead of being diverged off, they were blunted down, by vigorous imputations on either side, in order to diminish their effect. Any outliers on the environmental data can be related to actual ecological stress and any such information can be significant patterns erased.

Outcome of data preparation. After these steps the training and test data sets were totally imputed, scaled, encoded and eliminated redundant or irrelevant data. This gave it preprocessing and reproducibility of modeling phases.

Classification

They were classified in order to select the most effective models to predict vegetation condition.

Class imbalance

The issue of imbalance was perfectly shown through the counting of frequency of classes in which the At Risk was represented in significantly less frequencies than Degraded. To address this:

- Oversampled training data was used to train the models.
- The initial lopsided distribution was cross-validated and the approximations of realistic performance were gained.

This model learning design traded off model learning and fair evaluation.

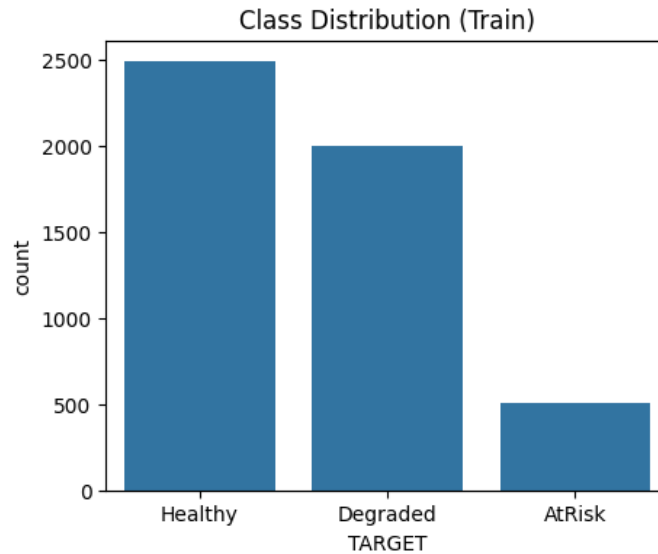


Figure 3: Class distribution

Model training and tuning

1. k-Nearest Neighbors (k-NN)

- **Hyperparameters tuned:**
 - Number of neighbors: {3, 5, 7}
 - Weighting scheme: {uniform, distance}
- **Chosen:** k=5 distance weighted.
- **Rationale:** Too many run the risk of overfitting the minority patterns; too few, too few are too few. Distance weighting causes the samples which are nearer to be considered more.

2. Naive Bayes (GaussianNB)

- **Hyperparameters that were optimized:** var smoothing {1e-9, 1e-8}.
- **Selected:** 1e-9.
- **Rationale:** NB is technically simple and fast, whereas the independence of features is assumed and it was not met in this instance due to the fact that the ecological variables were correlated with each other (Kuhn and Johnson, 2013). There was thus poor performance.

3. Decision Tree (DT)

- **Hyperparameters tuned:** max_depth {None, 5, 10}, min_samples_split {2, 5}.
- **Selected:** depth=10, min_samples_split=2.
- **Rationale:** Depth limitation prevents overfitting through reducing the complexity of the trees, but DTs were still not as stable as ensembles.

4. Random Forest (RF)

- **Hyperparameters tuned:** n_estimators {100, 200}, max_depth {None, 10}.
- **Selected:** 200 trees, depth=None.
- **Rationale:** Larger ensembles stabilize performance, and RF has demonstrated strong generalization in heterogeneous data settings (Breiman, 2001).

Validation scheme

The models were trained on 80 percent of the training data and tested on the 20 percent of the data. Moreover, the training split (5-fold) was cross-validated in order to forecast the generalization output. One indicator was accuracy and the other one was balanced accuracy and macro F1.

Model	Actual Accuracy (Validation)	Estimated Accuracy (CV)
k-NN	0.863	0.904
Naive Bayes	0.158	0.367
Decision Tree	0.767	0.905
Random Forest	0.906	0.973

Table 1: Model Comparison based on accuracy and estimated accuracy

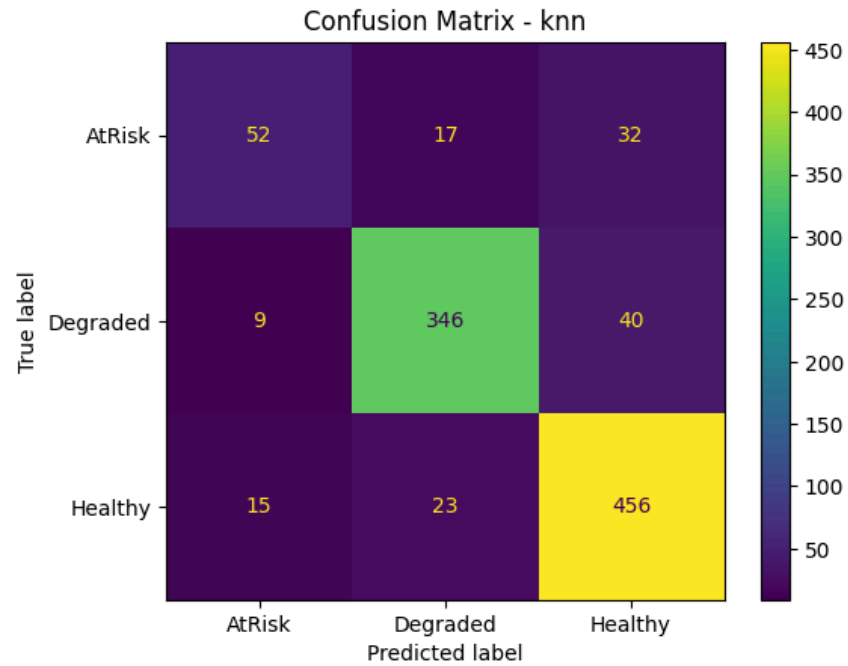


Figure 4: Confusion Matrix for k-NN

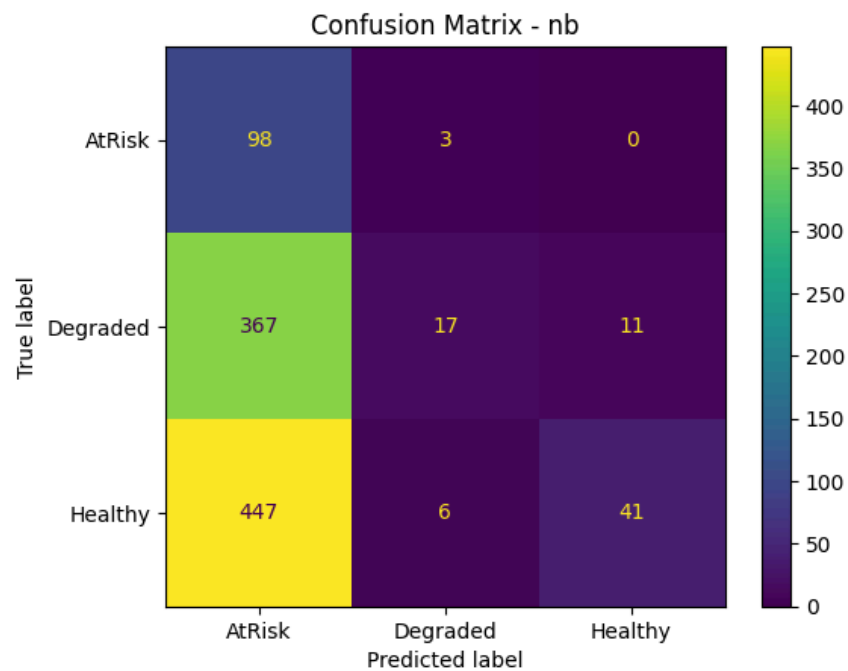


Figure 5: Confusion Matrix for Naive Bayes

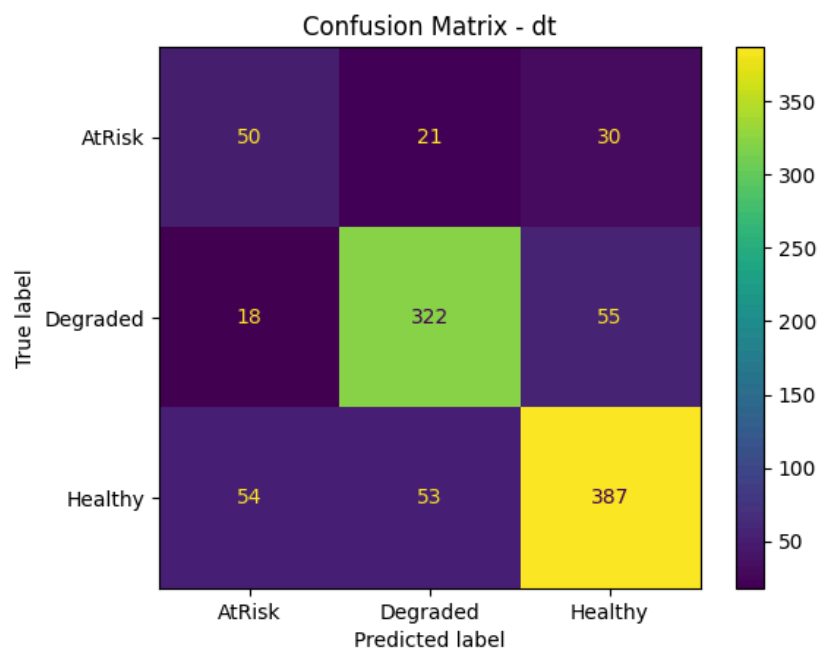


Figure 6: Confusion Matrix for Decision Tree

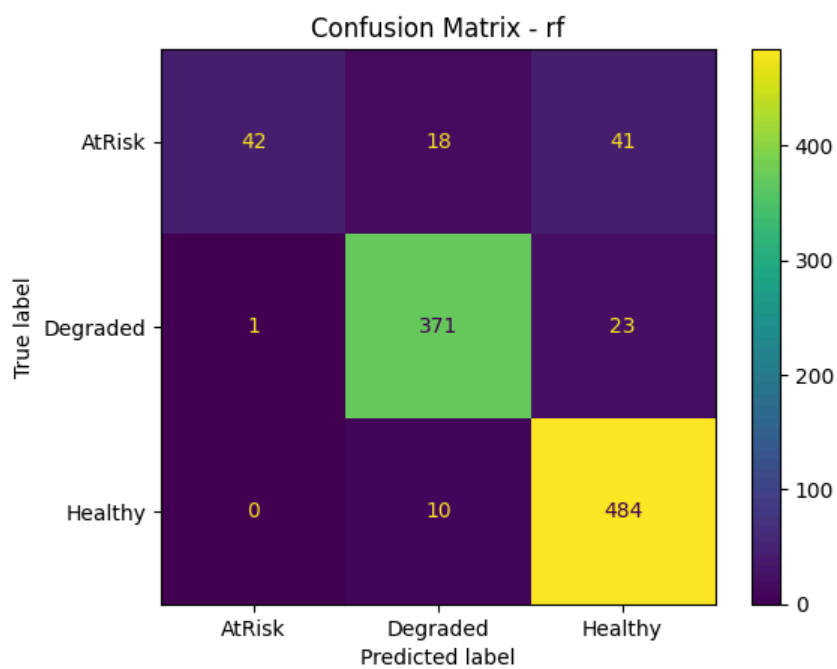


Figure 7: Confusion Matrix for Random Forest

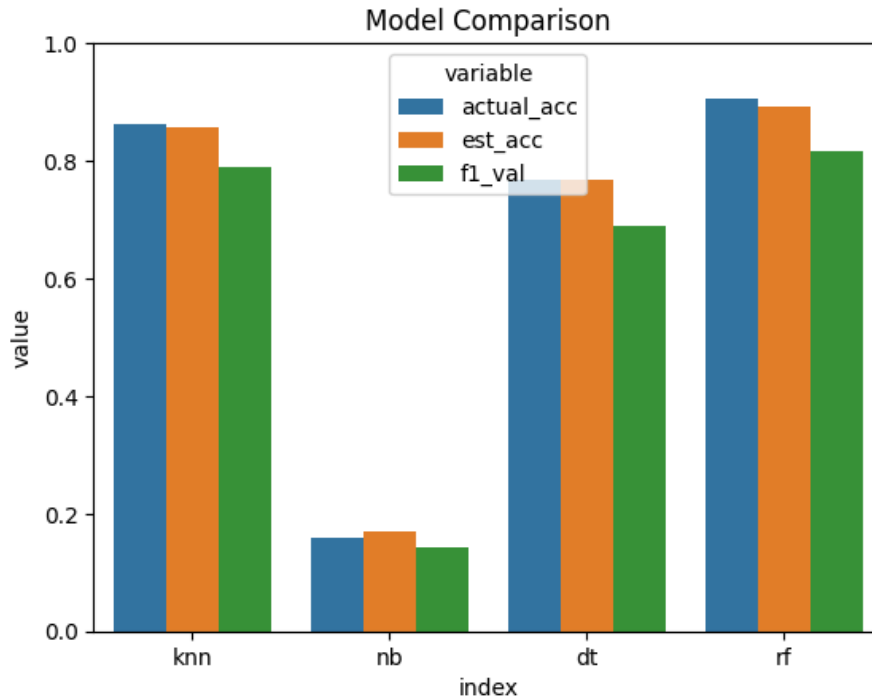


Figure 8: Model comparison

Discussion of results

- Random Forest was better in comparison with the remaining models and also its validation and macro-f1 accuracy were high. This is because of its high performance that is credited by its resistance to noise as well as its ability to sustain mixed type of data (Breiman, 2001).
- k-NN was scaling competitive and it also performed well. Nimble vegetation condition variations were determined by its local decision making.
- There was overfitting in Decision Tree because a large change in accuracy was observed between CV and validation accuracy.
- Naive Bayes did not work well due to the correlated features as it is expected to (Kuhn and Johnson, 2013).

Such findings may be explained by theoretical observations: ensemble methods are more reliable, and simple probabilistic methods fail in correlated features spaces.

Prediction

Models were then compared and random Forest and k-NN were selected as the best models. The two were re-trained with the full 5,000 training rows and transferred to 500 test unlabeled rows. Predictions on output were in the CSV format and 2 predictions were in a row, i.e.:

index	Predict1	Predict2
5000	Degraded	Degraded
5001	Degraded	Degraded
5002	Healthy	Healthy
.....		

Table 2: Visual representation of first few rows of the generated csv file

The accuracy of the two chosen models was reasonably estimated in relation to their validation outcomes, which implies that both the models can be assumed to provide consistent predictive performance in the case they are applied to the hidden test data. estimated accuracies of the two selected models were reasonably close to their validation results, suggesting that both models can be expected to deliver reliable predictive performance when applied to the unseen test data.

Conclusion

This paper has demonstrated that end-to-end predictive models could be created to monitor vegetation condition. The main contributions were:

- Intensive data cleaning, missing data, data duplication, irrelevant data, data scaling, numerical encoding, and class imbalance.
- There were four classifiers that were compared and the strongest ones were Random Forest and k-NN.
- The validation design was done meticulously to ensure that it improved the learning process and made the estimates realistic.

Limitations:

- The cross-validation estimates of weak models were overstated by the oversampling.

- Correlated ecological variables were a challenge to Naive Bayes.
- Decision Tree also had higher chances of overfitting.

Future work:

- Test synthetic oversampling methods e.g. SMOTE to improve balance (Chawla et al., 2002).
- Can implement some boosting methods such as XGBoost in the direction of powerful ensembles.
- Market predictions with domain experts to check their predictions against Indigenous ecological knowledge.

References

Breiman, L. (2001). Random Forests. *Machine Learning*, 45(1), 5–32.

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Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). SMOTE: Synthetic Minority Over-sampling technique. *Journal of Artificial Intelligence Research*, 16,

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